MATHS MANTHAN
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 06 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If one zero of the polynomial $(a^2+9)x^2+13x+6a$ is reciprocal of the other, then the value of a is: [1]

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) 9

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let zeroes be α and $\frac{1}{\alpha}$. We know Vieta's product formula:

Formula Used: Product of zeroes $= \frac{C}{A} \implies \alpha \cdot \frac{1}{\alpha} = 1 \implies A = C$

Here $a^2+9=6a \implies a^2-6a+9=0 \implies (a-3)^2=0 \implies \boxed{a=3}$. [1M]

Q2. If the zeroes of the polynomial x^2-bx+c are two consecutive positive integers, then b^2-4c is: [1]

(A) 1 (B) 2 (C) -1 (D) 0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let the consecutive integer zeroes be $\alpha = n$ and $\beta = n+1$, so $|\alpha-\beta|=1$. We know the discriminant-root identity:

Formula Used: $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$

With $\alpha+\beta=b$ and $\alpha\beta=c$, we get $b^2-4c=(\pm1)^2=\boxed{1}$. [1M]

Q3. If one zero of $p(x) = ax^2 + bx + c$ is double the other, then the relation between a, b, c is: [1]

(A) $2b^2 = 9ac$ (B) $9b^2 = 2ac$ (C) $b^2 = 4ac$ (D) $2a^2 = 9bc$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Let the zeroes be α and 2α. We know Vieta's relations:

Formula Used: $\alpha + 2\alpha = -\frac{b}{a}, \qquad \alpha(2\alpha) = \frac{c}{a}$

Hence $3\alpha = -\frac{b}{a} \implies \alpha = -\frac{b}{3a}$. Substituting into $2\alpha^2 = \frac{c}{a}$:

$$2\left(-\frac{b}{3a}\right)^2 = \frac{c}{a} \implies \frac{2b^2}{9a^2} = \frac{c}{a} \implies \boxed{2b^2 = 9ac}. \qquad [1M]$$

Q4. The value of k for which the zeroes of the polynomial $p(x) = 9x^2 - 12x + k$ are equal, is: [1]

(A) 4 (B) -4 (C) 3 (D) 2

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

We know that equal zeroes occur if and only if the discriminant vanishes:

Formula Used: $D = b^2 - 4ac = 0$

With $a = 9$, $b = -12$, $c = k$:

$$(-12)^2 - 4(9)(k) = 0 \implies 144 - 36k = 0 \implies k = \frac{144}{36} = \boxed{4}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): If α, β are zeroes of $kx^2 + 4x + 4$ such that $(\alpha + \beta)^2 - 2\alpha\beta = 24$, then $k = 1$ is an integer solution.

Reason (R): For $kx^2 + 4x + 4$, the sum of zeroes is $-\frac{4}{k}$ and the product of zeroes is $\frac{4}{k}$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (D) A is false, R is true.

We know Vieta's relations for $kx^2 + 4x + 4$ ($k \neq 0$):

Formula Used: $\alpha + \beta = -\frac{4}{k}, \qquad \alpha\beta = \frac{4}{k}$

Reason **(R)** correctly states $\alpha + \beta = -\frac{4}{k}$ and $\alpha\beta = \frac{4}{k}$, so **(R)** is **TRUE**. [0.5M]

Substituting into the condition in **(A)**:

$$\left(-\frac{4}{k}\right)^2 - 2\left(\frac{4}{k}\right) = 24 \implies \frac{16}{k^2} - \frac{8}{k} = 24 \implies 3k^2 + k - 2 = 0$$

$$\implies (3k - 2)(k + 1) = 0 \implies k = \frac{2}{3} \text{ or } k = -1.$$

Indeed, $k = 1$ gives $16 - 8 = 8 \neq 24$ (the actual integer solution is $k = -1$). Hence **(A)** is **FALSE** and **(R)** is **TRUE**, so option $\boxed{\text{(D)}}$ is correct. [0.5M]

Q6. Find the condition on the coefficients of $ax^2 + bx + c$ such that its zeroes are equal in magnitude but opposite in sign. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the two zeroes be α and $-\alpha$ ($a \neq 0$). We know Vieta's sum and product identities:

Formula Used: Sum of zeroes $= -\frac{b}{a}$, Product of zeroes $= \frac{c}{a}$

Equating the sum of zeroes to $-\frac{b}{a}$:

$$\alpha + (-\alpha) = -\frac{b}{a} \implies 0 = -\frac{b}{a} \implies \boxed{b = 0} \quad (\text{since } a \neq 0). \qquad \textbf{[1.5M]}$$

Furthermore, for distinct real opposite zeroes ($\alpha \neq 0$), $\alpha(-\alpha) = -\alpha^2 = \frac{c}{a} < 0$, i.e., a and c have opposite signs ($ac < 0$). **[0.5M]**

Q7. If α, β are the zeroes of the polynomial $3x^2 - 2x - 1$, obtain a quadratic polynomial whose zeroes are $\alpha + 2\beta$ and $2\alpha + \beta$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $3x^2 - 2x - 1$, we have $\alpha + \beta = \frac{2}{3}$ and $\alpha\beta = -\frac{1}{3}$. We know:

Formula Used: $p(x) = k\left[x^2 - S'x + P'\right]$, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Computing the sum S' and product P' of the new zeroes $\alpha' = \alpha + 2\beta$ and $\beta' = 2\alpha + \beta$:

$$S' = (\alpha + 2\beta) + (2\alpha + \beta) = 3(\alpha + \beta) = 3\left(\frac{2}{3}\right) = 2, \qquad \textbf{[1M]}$$

$$P' = (\alpha + 2\beta)(2\alpha + \beta) = 2(\alpha^2 + \beta^2) + 5\alpha\beta = 2\left[(\alpha + \beta)^2 - 2\alpha\beta\right] + 5\alpha\beta$$

$$= 2(\alpha + \beta)^2 + \alpha\beta = 2\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right) = \frac{8}{9} - \frac{3}{9} = \frac{5}{9}. \qquad \textbf{[1M]}$$

Thus $p(x) = k\left(x^2 - 2x + \frac{5}{9}\right)$; taking $k = 9$ gives $\boxed{p(x) = 9x^2 - 18x + 5}$. **[1M]**

Q8. If α and β are the zeroes of the quadratic polynomial $2x^2 + 3x + 1$, obtain a quadratic polynomial whose zeroes are $\alpha - 2\beta$ and $\beta - 2\alpha$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $2x^2 + 3x + 1$, we have $\alpha + \beta = -\frac{3}{2}$ and $\alpha\beta = \frac{1}{2}$. We know:

Formula Used: $p(x) = k\left[x^2 - S'x + P'\right]$, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

For the new zeroes $\alpha' = \alpha - 2\beta$ and $\beta' = \beta - 2\alpha$:

$$S' = (\alpha - 2\beta) + (\beta - 2\alpha) = -(\alpha + \beta) = -\left(-\frac{3}{2}\right) = \frac{3}{2}, \qquad \textbf{[1M]}$$

$$P' = (\alpha - 2\beta)(\beta - 2\alpha) = \alpha\beta - 2\alpha^2 - 2\beta^2 + 4\alpha\beta = 5\alpha\beta - 2\left[(\alpha + \beta)^2 - 2\alpha\beta\right]$$

$$= 9\alpha\beta - 2(\alpha + \beta)^2 = 9\left(\frac{1}{2}\right) - 2\left(-\frac{3}{2}\right)^2 = \frac{9}{2} - \frac{9}{2} = 0. \qquad \textbf{[1M]}$$

Hence $p(x) = k\left(x^2 - \frac{3}{2}x\right)$; setting $k = 2$ gives $\boxed{p(x) = 2x^2 - 3x}$. **[1M]**

Q9. Find the zeroes of the polynomial $v(x) = v^2 + 4\sqrt{3}\,v - 15$ by factorisation and verify the relationship between zeroes and coefficients. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Treating the quadratic expression in variable v, $q(v) = v^2 + 4\sqrt{3}\,v - 15$, we have $a = 1$, $b = 4\sqrt{3}$, $c = -15$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Splitting $4\sqrt{3} = 5\sqrt{3} - \sqrt{3}$ (since $(5\sqrt{3})(-\sqrt{3}) = -15$):

$$v^2 + 5\sqrt{3}\,v - \sqrt{3}\,v - 15 = v(v + 5\sqrt{3}) - \sqrt{3}(v + 5\sqrt{3}) = (v + 5\sqrt{3})(v - \sqrt{3}). \qquad \textbf{[1M]}$$

Setting to 0 yields zeroes $\alpha = \boxed{-5\sqrt{3}}$ and $\beta = \boxed{\sqrt{3}}$. **[1M]**

Step 2 – Verification:

$$\alpha + \beta = -5\sqrt{3} + \sqrt{3} = -4\sqrt{3}, \qquad -\frac{b}{a} = -\frac{4\sqrt{3}}{1} = -4\sqrt{3}, \qquad \textbf{[0.5M]}$$

$$\alpha\beta = (-5\sqrt{3})(\sqrt{3}) = -15, \qquad \frac{c}{a} = \frac{-15}{1} = -15. \qquad \textbf{[0.5M]}$$

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Q10. Case Study: Botanical Garden Water Fountain

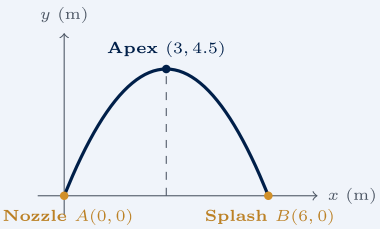
A water fountain shoots a jet from nozzle A ($x = 0$) into a pond at B, modelled by $y = -\frac{1}{2}x^2 + 3x$, where x is horizontal distance (m) and y is water height (m). [4]

(i) Find the horizontal distance between the nozzle A and the splash point B. **[1]**

(ii) What is the maximum height attained by the water stream above the pond level? **[1]**

(iii) **(a)** At what horizontal distance(s) from nozzle A is the water stream at a height of 4 m? **OR (b)** If the jet is adjusted to $y = -x^2 + 4x$, calculate its new maximum height and horizontal range. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]



Hydraulic Parabolic Path: The stream starts at nozzle $A(0,0)$, reaches apex $(3, 4.5)$ by completing the square, and splashes onto pond surface at $B(6,0)$.

Formula Used: $y = -\frac{1}{2}x^2 + 3x = -\frac{1}{2}(x-3)^2 + 4.5$

(i) Horizontal range AB: Setting $y = 0$:

$$-\frac{1}{2}x^2 + 3x = 0 \implies -\frac{1}{2}x(x-6) = 0 \implies x = 0 \text{ (at } A) \text{ or } x = 6 \text{ (at } B).$$

Hence distance $AB = 6 - 0 = \boxed{6 \text{ m}}$. [1M]

(ii) Maximum height: Completing the square:

$$y = -\frac{1}{2}(x^2 - 6x) = -\frac{1}{2}[(x-3)^2 - 9]$$
$$= -\frac{1}{2}(x-3)^2 + \frac{9}{2} = -\frac{1}{2}(x-3)^2 + 4.5.$$

Since $-\frac{1}{2}(x-3)^2 \le 0$, the maximum height is $\boxed{4.5 \text{ m}}$ (at $x = 3$ m). [1M]

(iii)(a) Distance(s) where $y = 4$ m:

$$-\frac{1}{2}x^2 + 3x = 4 \implies x^2 - 6x + 8 = 0$$
$$\implies (x-2)(x-4) = 0 \implies \boxed{x = 2 \text{ m and } x = 4 \text{ m}}. \quad [2M]$$

OR (Alternative Part (b) – Adjusted jet $y = -x^2 + 4x$):

$$\text{Range } (y = 0): \quad -x^2 + 4x = 0 \implies x = 0,\ 4 \implies \boxed{4 \text{ m}}, \quad [1M]$$
$$\text{Max height}: \quad y = -(x-2)^2 + 4 \le 4 \implies \boxed{4 \text{ m}}. \quad [1M]$$

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